

PAVS: Bridging AMD's Softpower Gap

From the Voice AI Perspective

Why a competitive silicon doesn't win by itself?

Voice AI is the lens -- conclusions Hold

The Voice AI: Business, Opportunities & Tech-Stack

A Massive Inference Market, Being Decided Now — Next is Edge Inference

Tech Stack

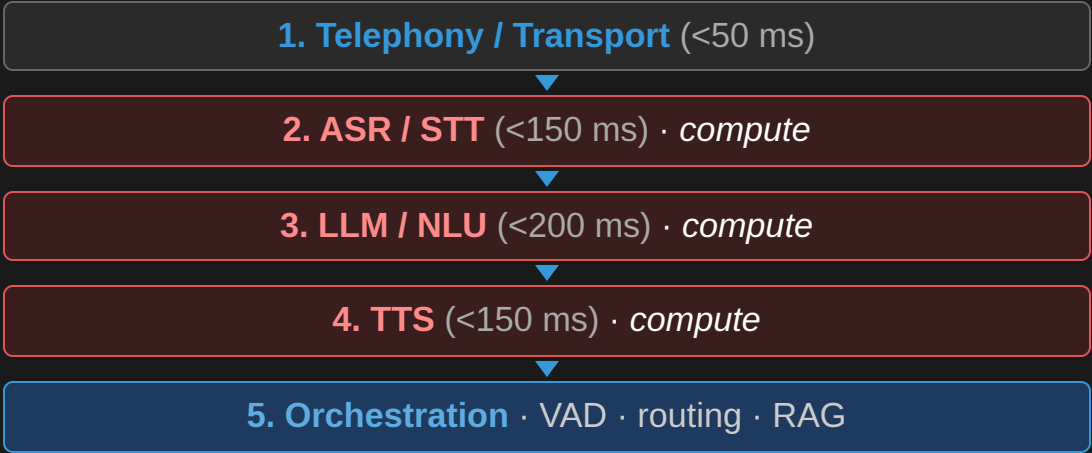


Fig 1 · 3/5 layers are GPU-compute-bearing

— AMD's advantage *and* its gaps both live here.

The opportunity

Metric	Value
AI voice-agents market, 2034	\$47.5B @ 35% CAGR
Cost per call — human vs AI	\$7–12 → ~\$0.40
YoY growth, production deploys	340% (500+ orgs)
Contact-center labor savings, 2026	\$80B (Gartner)
Orgs planning voice AI by end-2026	80%

A massive, fast-growing inference market whose compute default is being chosen right now.

It's a Compute Market — AMD Has the Winning Silicon

A deployed agent is a steady-state inference machine

- It doesn't train — it **transcribes, reasons & speaks 24×7**
- **LLM** is cheapest per minute but **most latency-critical** — first token in **<200 ms**
- **TTS is the hidden budget killer** — premium hosted synthesis is **\$0.10–\$0.30/min**, often exceeding ASR + LLM combined
- The next frontier is **on-device / edge inference** — kiosks, vehicles, factory floors, hospitals and public services

The savings compound at scale

- **34% lower cost-per-token** → **~\$84K/month** saved at 1M min/month
- Self-hosted ROCm TTS removes **\$100K–\$300K/month** in hosted synthesis
- on-prem: **~50% lower GPU CapEx**

Dimension	MI300X	H100 SXM5
VRAM	192 GB HBM3	80 GB HBM3e
On-demand cloud price	\$1.50–2.50/hr	\$2.90/hr
Throughput (tok/s)	~18,000	~19,500
Cost / 1M tokens	\$0.027	\$0.041
70B fits on 1 GPU?	Yes	No (needs 2+NVLink)

LLM inference — MI300X vs H100 SXM5 (Llama 3.1 70B, 2026)
MI355X (288 GB HBM3e) lands within single-digit % of B200 on
MLPerf Inference v6.0.

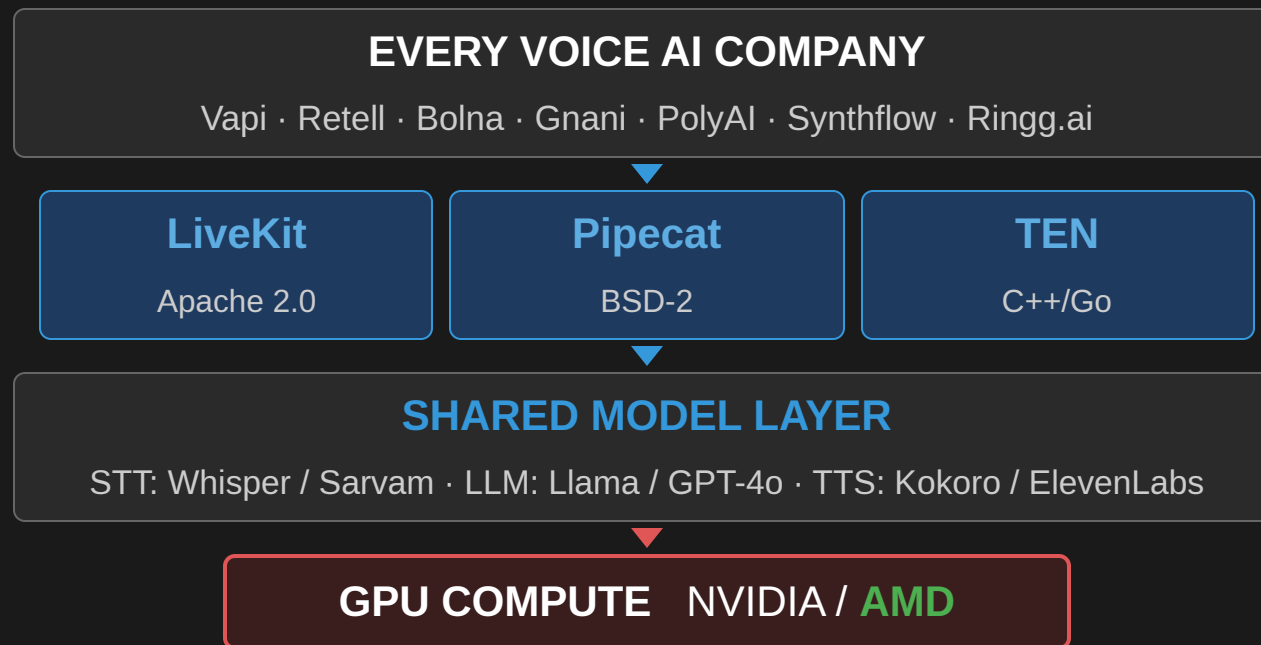
¹ Spheron: ROCm vs CUDA GPU Cloud 2026 — cloud pricing MI300X vs H100

² vLLM Blog: High-Performance Inference on AMD ROCm — throughput benchmarks

Silicon advantage is yet unrealized in the Market: **This is not a hardware problem.**

Voice AI Stack Convergence & One Compute Decision

*Despite hundreds of companies and billions in funding, production code sits above just **three open-source frameworks** — sharing one model layer and one compute decision.*



Owning the compute isn't enough — **you win by owning the on-ramps just above it: HW choice is made before you reach the compute layer.**

How NVIDIA Became the Gateway — The NIM Playbook

*Dominance wasn't silicon alone. The instrument was **NIM** — pre-packaged containers that make NVIDIA the path of least resistance at the first developer touchpoint.*

Ready-to-run microservices

- **Riva ASR NIM** — streaming ASR (Parakeet) in one `docker pull` ; no config, no VAD tuning
- **Maxine Studio Voice NIM** — real-time noise suppression / echo cancellation → directly lowers ASR word-error-rate

Complete applications

- **AI Blueprints** — deployable voice-agent repos, not docs. The GPU choice is already encoded inside.

Developer & cloud on-ramps

- **RTX Spark** — Grace+Blackwell devkit, 128 GB shared memory; prototype on NVIDIA → default to NVIDIA in production
- **Azure AI Foundry NIM** (July 2026) — deploy from VS Code in one click, NVIDIA GPU underneath
- **SI certification** — TCS, Infosys, Wipro, Accenture, Capgemini trained & certified on NIM

You never consciously select a GPU — Choice is already made. - NVIDIA is the default; every other choice requires active effort.
That is softpower.

CUDA Gap Closed; Ecosystem gap exists

ROCm support for voice-AI components (2026)

Component	ROCm	Note
LLM serving (vLLM, Llama 3.1)	Full	90–95% of H100 tput
LLM serving (SGLang)	Full	Within 5–10% of CUDA
TTS: Kokoro, Chatterbox	Full	Pure PyTorch, 0 changes
TTS: Coqui XTTS-v2	Partial	Works with effort
ASR: Faster-Whisper (batch)	Full	RT mode needs work
ASR: streaming containers	Gap	CUDA-assumed
Embeddings: HF TEI	Gap	No ROCm build

NIM ecosystem gap: NVIDIA vs AMD required

NVIDIA Has	AMD Equivalent	Status
Riva ASR NIM	AMD ASR container	Absent
Riva TTS NIM	AMD TTS container	Absent
AI Blueprint: voice agent	Pipecat on MI300X	Absent
Pipecat NVIDIA processor	Pipecat AMD processor	Absent
LiveKit certified backend	LiveKit AMD backend	Absent
VS Code NIM extension	ROCm / Quark ext.	Absent
Azure Foundry NIM	Azure AMD endpoint	Partial
SI cert (TCS, Infosys)	SI cert program	Absent
28M devs default CUDA	AMD dev default	Opt-in

Framework ecosystem presence

Framework	NVIDIA	AMD
Pipecat	Official processor	Absent
LiveKit Agents	Certified backend	Absent
LangChain	langchain-nvidia-ai-endpoints	Absent
LlamaIndex	llama-index-llms-nvidia	Absent
Haystack	NIM pipeline component	Absent

The gap is not silicon — it's the **least-resistant path to silicon**: containers, plugins, blueprints & marketplace slots.

The Five Downstream Ecosystem Layers

Same logic at every layer: ship a named package → developers adopt the default → the hardware choice becomes invisible at the call site.

1 · Framework plugin slots

`ChatNVIDIA(...)` routes to NVIDIA at the endpoint. AMD doesn't exist at that call site — repeated across millions of pipelines.

2 · Inference serving infrastructure

Triton / Dynamo + TensorRT-LLM = the invisible substrate (20–30% edge at batch 1–4). AMD has ****vLLM ROCm only**** — one tool, not infrastructure.

3 · Voice-domain SDKs

Riva (ASR+TTS+MT+diarization) · Maxine (noise suppression) · ACE (digital humans).

****AMD has no equivalent for any of the three.****

4 · Edge hardware platform

Jetson (Nano → Thor) + JetPack ships Riva, vLLM, CUDA. Certified OEMs: Siemens, KUKA, ADLINK. Caterpillar runs Jetson Thor + Riva inside a bulldozer cab. AMD — **partial**: Kria (latest) launched but not ecosystem-certified for industrial voice pipelines.

5 · Cloud marketplace slots

NIM native on Azure Foundry; selectable on HF, SageMaker, Vertex. AMD **partial**: [`huggingface/testing-rocm7.0-preview`](https://hub.docker.com/r/huggingface/testing-rocm7.0-preview) exists — preview only, ****not surfaced in the UI****.

The pattern: ship the default package, and the hardware choice disappears at the call site. AMD is not present above Compute layers .

The Four Upstream Gaps

Gap 1 · Training & fine-tuning moat

Voice companies fine-tune weekly (custom ASR/TTS/NLU). **NeMo → Triton export** locks the whole loop to NVIDIA.

Gap 2 · ISV & SI sales channel

GPU chosen implicitly by the ISV. NVIDIA is in **Genesys** (11k+ customers), **Cisco Webex** (Maxine), **Salesforce**, **ServiceNow**. SIs (TCS, Infosys, Wipro, Accenture) are **certified on NVIDIA**. AMD: no presence in any major contact-center ISV.

Gap 3 · Voice + vision convergence

Next-gen agents fuse camera + mic. Jetson Thor unifies **Isaac · Metropolis · Riva · ACE** on one substrate. AMD's lines still converging **AMD-PAVS AI Pipelines** — a **converging but early stage**.

Gap 4 · Synthetic data for low-resource languages

No AMD-native synthetic-data pipeline's or minimal academic partnership's → lifecycle dependency predates any inference decision.

Closing downstream puts AMD in contention; closing upstream creates platform momentum.

The AMD Gateway Strategy

Run NVIDIA's playbook — targeting the cost wedge its pricing creates. Same Pipecat code, same LiveKit transport, 34% lower per-conversation LLM cost.

Step 1 · Three certified containers

``docker pull`` → validated in 30 s: (a) Pipecat Voice Agent (Whisper + vLLM + Kokoro), (b) LiveKit + SIP, (c) DPDP-compliant, on-prem setup.

Step 2 · AMD Voice AI Blueprints

Deployable repos, not docs — US/EU contact center, India BFSI, Arabic GCC, edge/offline. Must run on a ****~\$700 consumer GPU****.

Step 3 · Pipecat ROCm plugin

``pipecat-ai[amd]`` on PyPI — AMD as an official processor alongside Deepgram / ElevenLabs.

Step 4 · HF TEI on ROCm

Upstream ROCm into Text-Embeddings-Inference — closes the last RAG software gap.

Step 5 · HF Inference Endpoints slot

TGI ROCm is ready — one ****business agreement**** makes MI300X selectable. ****Zero eng weeks.****

Step 6 · Ryzen AI NPU on-ramp

``pipecat create --backend amd-npu`` — offline, no GPU, no API key. The dev-laptop demo that mirrors ****RTX Spark****.

PAVS sits exactly at this intersection — between AIG model outputs and lighthouse customers. It does **Implicitly** what NVIDIA's platform teams do intentionally

Prioritized Action Roadmap

Tier 1 — Pure engineering (no new silicon, no product decisions)

Action	Effort	Horizon	Impact
HF Inference Endpoints slot	Business neg.	0–4 wks	High
Certified ASR + TTS containers	2 wks each	1 month	High
Pipecat AMD processor (PyPI)	2 wks	1 month	Medium
LangChain / LlamaIndex / Haystack	2–4 wks ea.	1–2 mo	High
AMD Voice AI Blueprints (4 targets)	4 wks each	2 months	High
<code>bitsandbytes</code> ROCm official	4–6 wks	2 months	High
Triton ROCm certified image	12–16 wks	4 months	High
Ryzen AI NPU voice template	4 wks	2 months	Medium

Tier 2 — Business dev + engineering

Action	Effort	Horizon	Impact
Genesys ISV integration	Business dev	6–12 mo	Very high
SI cert program (TCS, Infosys)	Program build	6–12 mo	Very high

First 12–16 weeks: Steps 1–4 with 6–8 engineers make AMD visible at first contact. **India BFSI under DPDP is the natural beachhead** — on-prem mandatory, ~50% lower CapEx.

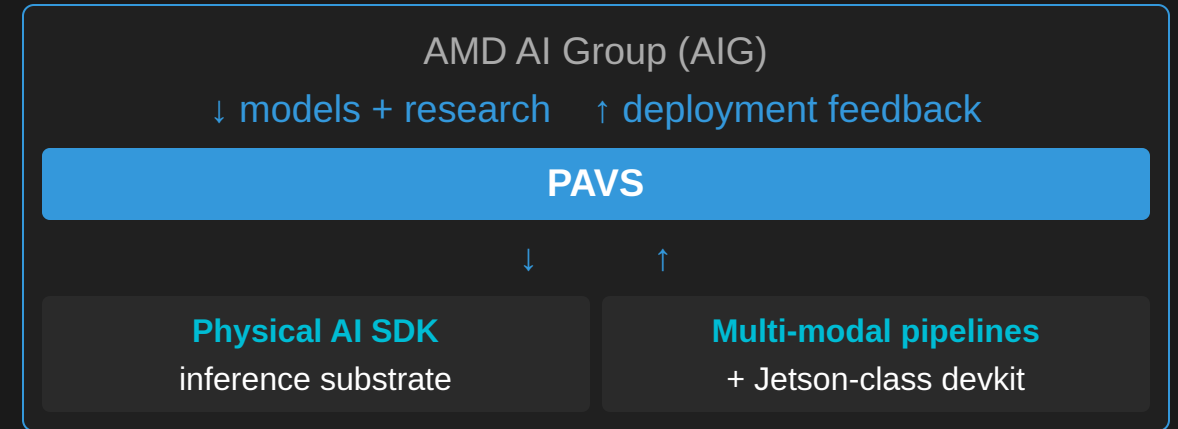
How the Bridge Is Being Closed

The gap can't be closed by a new chip

- It's a **software & ecosystem** problem — containers, plugins, blueprints, marketplace slots
- Which is **exactly** the problem a **vertical-software team** solves

PAVS produces the closing assets organically

- Each lighthouse deployment → a **reusable pipeline** = a step toward an AMD-native ecosystem
- Each model the Physical AI SDK supports → **one less** requiring NVIDIA tooling on AMD HW
- Each robotics pipeline on Ryzen AI APU → a **reference design** for an OEM / SI / ISV



Make AMD the path of least resistance at the first touchpoint, and the rest of the stack follows.